usb-buck — Design Document

PCB-0002-A

hwde v1 pipeline

generated 2026-09-25 18:07:05

Contents

1 Board and Run Metadata **1**

2 Overview **1**

3 Requirements **2**

4 Architecture **4**

5 Schematic **8**

6 Layout **14**

7 Verification **15**

8 DFM and Fabrication **18**

9 Run Record (Appendix) **19**

10 Artifact Index **22**

1 Board and Run Metadata

Field	Value
board	usb-buck
part number	PCB-0002-A
workspace	`boards/PCB-0002-A_usb-buck`
phase	P10
gate status	all 5 recorded gates pass
state created	2026-07-28T01:26:50
state updated	2026-08-14T11:25:39

2 Overview

Brief: usb-buck (S14 run b)

A 4-layer STM32 USB device board:

- STM32F103C8T6 MCU, 8 MHz crystal, SWD header (4-pin)
- USB micro-B connector, USB 2.0 full-speed DEVICE on the MCU's USB pins
- Powered from USB VBUS (5V) through an AP63203 buck converter to 3.3V
- One status LED on a GPIO; one user button
- JLCPCB 4-layer, economy PCBA, max 55x45 mm, prototype qty 10

- Prefer JLC Basic parts; hand-solderable connectors acceptable
- Structure the schematic hierarchically (e.g. power / mcu / usb sheets) unless the architect judges flat clearly better

Fabrication spec snapshot

Key	Value
layers	4
width_mm	50.0
height_mm	40.0
qty	10
surface_finish	HASL
solder_mask_color	green
assembly	True

Stackup

Chosen stackup: `JLC04161H-3313` (board class: 4-layer, 1.6 mm)

3 Requirements

Requirements: usb-buck

1. Function

USB-powered STM32 device board. STM32F103C8T6 MCU (clocked from an 8 MHz crystal) runs USB 2.0 full-speed device firmware, talking to a host over a micro-B USB connector that also supplies board power. Onboard status LED and user button give minimal local UI/debug feedback; a 4-pin SWD header supports programming and debugging. Compact prototype/dev board, build qty 10. Note: brief requests a hierarchical schematic (e.g. power/mcu/usb sheets) unless the architect judges a flat schematic clearly better - carried forward as a directive for the architecture stage, not a board requirement.

2. Interfaces

- USB: USB 2.0 full-speed DEVICE (stated). Connector: micro-B (stated). D+/D- wired to the MCU's fixed USB pins - PA11 (D-) / PA12 (D+) on the STM32F103C8T6 (verified against part datasheet references). Device only: no host, hub, or OTG function.
- SWD: 4-pin header for programming/debug (stated). ASSUMED: standard minimal SWD-only pinout - SWCLK, SWDIO, GND, 3V3 reference (no NRST, no SWO); 2.54 mm pitch, unshrouded pin header. Low risk - architect picks exact pin order and silkscreen labels it.
- User I/O (onboard, no external connector):
  - 1x status LED, GPIO-driven. ASSUMED: color/pin is architect's choice, cosmetic only.
  - 1x user button, GPIO input. ASSUMED: generic application input (not BOOT0/bootloader select, not NRST/reset) - brief calls it "user button" distinctly from any programming function.

3. Power

- Source: USB VBUS only. No battery, no charging, no mains input.
- Input: 5V nominal (USB spec range ˜4.75-5.25V). Budget: 500mA-class (standard USB downstream port; no PD/QC negotiation implied or needed).
- Regulation: AP63203 buck converter, VBUS (5V) -> 3.3V rail (stated, named part). NOTE: this explicit part name overrides the general "prefer JLC Basic parts" preference for this one IC; architect verifies sourcing (JLC Basic/Extended/LCSC) and applies the Basic-parts preference to every other part.
- Rail budget (GUESS, for architect sizing only): MCU active ˜50mA, LED ˜5-10mA, misc pull-ups/passives ˜5mA -> well under 150mA on the 3.3V rail. Comfortably inside both the buck converter's and the USB port's capability.

4. Environment

Not stated. ASSUMED: indoor bench/lab prototype use, ordinary room ambient (roughly 0-40C), no enclosure, no ingress protection, no vibration/shock requirement. Low risk - qty-10 prototype board, easy to revisit.

5. Size & mounting

- Outline: max 55x45 mm (stated).
- Mounting holes: not stated. ASSUMED: none required (bare board, no enclosure per Environment).
- Height limit: not stated. ASSUMED: no restriction (open board, no enclosure).

6. Quantity & budget

- Build quantity: prototype qty 10 (stated). ASSUMED: one-off prototype run - no follow-on production volume implied or planned for.
- Target unit cost: no number stated. ASSUMED: no hard cap; cost is minimized via the stated strategy (JLC Basic parts preference + JLCPCB economy PCBA tier) rather than a fixed target.

7. Assembly

- JLCPCB PCBA, economy tier (stated).
- Parts: prefer JLC Basic library (stated), except AP63203 which is explicitly named (see Power note).
- Connectors: "hand-solderable connectors acceptable" (stated). ASSUMED interpretation: USB micro-B and/or the SWD header may be hand-soldered post-PCBA instead of placed by JLC assembly, if that is cheaper or simplifies sourcing; architect decides per part availability.
- Sidedness: not stated. ASSUMED: single-sided (top-only) SMT assembly - small part count and board size make this the simpler/cheaper default; architect may move to double-sided if placement requires it.

8. Compliance/safety flags

None apply:

- Mains voltage: no - USB-powered only, 5V max input.
- Batteries: no.
- Motors: no.
- >30V: no - max voltage on board is 5V (VBUS), stepped down to 3.3V.
- High current (>3A): no - USB 500mA-class budget; actual estimated draw well under 150mA.
- RF transmit: no - USB device function only, no radio.

9. Open questions

1. Target unit cost: is there a hard per-unit cost cap, or is "minimize via JLC Basic parts + economy PCBA" (the stated strategy) sufficient without a specific number? DEFAULT: no hard cap; stated strategy stands. 2. Enclosure / mounting: will this board be mounted in an enclosure or chassis (needing mounting holes and/or a height limit), or is it a bare bench-use board? DEFAULT: bare board, no mounting holes, no height limit.

Answers (P0 batch, AUTO-approved defaults per user directive)

1. Unit cost: no hard cap; JLC Basic + economy PCBA strategy stands. 2. Bare bench-use board: no mounting holes, no height limit.

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	S14 exec plan applies: auto-approve checkpoints w/ recommended defaults, model-mix agents, lean inline script phases	User directives 2026-07-27/28
P0	P0 defaults: no cost cap; bare board no mounting holes	AUTO per user directive; both low-risk defaults
P2	JLC04161H-3313 4L standard (not impedance-controlled - FS does not need it); In1 GND, In2 +3V3; 3-sheet hierarchy usb/power/mcu, net names contractual (/USB_DP,/USB_DM root-labeled; /mcu/OSC_* couple constraints to sheet name - deliberate)	Architect P2, research-grounded, hierdemo-verified naming
P2	Settled: buck stands; LED 1k on PC13 (3mA limit); DP pull-up hardwired 1.5k; VBUS sense OMIT; shell->GND direct; VBUS caps exactly 10uF+100nF (USB inrush ceiling == buck C_IN)	All 7 research OPEN items closed with reasons in decisions.md

Phase	What	Why
P3	Inductor L1 DCR 60mOhm vs DS43698's <30mOhm recommendation: WAIVED - the <30 criterion targets 2A-class operation; this board peaks 57mA (0.2mW conduction loss). Isat 2.9A >> 1.35x requirement holds	Engineering waiver with math; schematic-reviewer may contest
P3	ONE schematic-block agent owns all 3 sheets + root (vs one-per-sheet guidance)	28-part board; hierarchy fully preserved in the OUTPUT; single agent is cheaper and avoids cross-agent refdes/net coordination risk at this size
P3	Library edits approved: silk sanitize on C0603/C0805/R0603/LED (sub-minimum dots on pad 1) + U2 TSOT-26 gains a printable pin-1 dot; no lib_pull re-runs (easyeda2kicad dup bug + API 403)	Intra-footprint silk would hard-fail P5 self-check by design; U2 rotation = dead board
P5	Outline fixed 50x40 (45x35 failed: shelf pack needs the room); 40x30 target unreachable at v1 (no outline-shrink step)	Run (a) precedent; cap 55x45 respected
P7	USB pair max_uncoupled_mm 5.0 -> 8.0: measured 6.61mm decomposes to 3.6mm TVS flow-through package span + 2.9mm D+ pull-up stub, ZERO avoidable meander; FS critical length ~115mm makes this cosmetic	Structural spec correction, not a waiver of real coupling debt; researcher's 5.0 was a generic FS default

Sheet plan

usb-buck - hierarchical sheet plan

Three child sheets under a thin root, per the brief's request for a hierarchical schematic. The board has ~28 components across three clearly separable domains and exactly TWO nets that cross a sheet boundary, so hierarchy costs almost nothing in net-name risk and buys parallel P4 sheet generation.

The sheet NAMES below are contractual. They appear verbatim in netlist net names (`/mcu/OSC_IN`) and `constraints.json` already references them; renaming a sheet renames nets and breaks `netlist_audit --constraints`.

```
| Sheet | File | Blocks | Interface nets (sheet pins) | pwr_base |
|---|---|---|---|---|
| root ‘usb-buck‘ | ‘usb-buck.kicad_sch‘ | none (stitching only) | - | 1 (unused)
|
| ‘usb‘ | ‘usb.kicad_sch‘ | USB port + ESD | ‘USB_DP‘, ‘USB_DM‘ | 300 |
| ‘power‘ | ‘power.kicad_sch‘ | buck regulation | **none** | 100 |
| ‘mcu‘ | ‘mcu.kicad_sch‘ | MCU core + clock, user I/O, debug | ‘USB_DP‘, ‘USB_DM‘
| 200 |
```

1. Net naming - the CANONICAL final netlist names

Mechanism (verified in this repo: `tests/s7_regen/hierdemo/kicad/hierdemo.net` produces `+3V3`, `GND`, `/VIN`, `/CTL`, `/load/LED_K`):

1. **Power SYMBOLS make a net global across the whole hierarchy** with a BARE name and need NO sheet pin. 2. A child net merged with the root through a sheet pin takes the **root-side

label**: `schlib.Project.add_sheet(child, ..., nets=["X"])` wires each sheet pin outward to a root local label `X`, so the net becomes `/X`. 3. A child-internal label becomes `/<sheet>/NAME`.

Global rails (power symbols, no sheet pins) - bare names

Net	Symbol	Present on	Driven by
‘VBUS‘	‘power:VBUS‘	usb, power	J1 pin 1 (passive) -> **needs PWR_FLAG**
‘+3V3‘	‘power:+3V3‘	power, mcu	L1 pin 2 (passive - the AP63203 has no VOUT pin, the rail forms after the inductor) -> **needs PWR_FLAG**
‘GND‘	‘power:GND‘	all	**needs PWR_FLAG**

All three PWR_FLAGs live on the `power` sheet (`#FLG0100+`). Power symbols being global, one flag each drives the net hierarchy-wide.

Root-crossed signals - `/NAME`

Final net	Root label	Exposed by	Members
‘/USB_DP‘	‘USB_DP‘	‘usb‘ + ‘mcu‘ sheet pins	J1 pin 3, U3 channel-A I/O pads, U1 PA12, R4
‘/USB_DM‘	‘USB_DM‘	‘usb‘ + ‘mcu‘ sheet pins	J1 pin 2, U3 channel-B I/O pads, U1 PA11

Both children expose the same two names; the root's two local labels per name merge by name on the root sheet. These are the names `constraints.json` uses for `high_speed` and `diff_pairs` - matching the usbbuck4 golden precedent (`/USB_DP` + `/USB_DM` with bare `VBUS`/`+3V3`/`GND`).

Sheet-internal signals - `/<sheet>/NAME`

Sheet	Final net	Members
power	‘/power/SW‘	U2 SW, L1 pin 1, C3 (BST cap) low side
power	‘/power/BST‘	U2 BST, C3 high side
mcu	‘/mcu/OSC_IN‘	U1 PD0/OSC_IN, Y1, C10
mcu	‘/mcu/OSC_OUT‘	U1 PD1/OSC_OUT, Y1, C11
mcu	‘/mcu/SWDIO‘	U1 PA13, J2 pin 4
mcu	‘/mcu/SWCLK‘	U1 PA14, J2 pin 2
mcu	‘/mcu/NRST‘	U1 NRST, C19
mcu	‘/mcu/BOOT0‘	U1 BOOT0, R3 (to GND)
mcu	‘/mcu/LED_A‘	R1 (from +3V3) -> D1 anode
mcu	‘/mcu/LED‘	D1 cathode -> U1 PC13 (active low sink)
mcu	‘/mcu/BTN‘	U1 PB0, R2 (to +3V3), SW1 (to GND)
usb	-	none: the pair and the rails are all cross-sheet or global

`constraints.json` references `/mcu/OSC_IN` and `/mcu/OSC_OUT` in `high_speed`. That is deliberate (see decisions.md item 8) and is the reason the sheet name `mcu` is contractual.

2. Refdes allocation (unique ACROSS sheets - contractual for P4)

Sheet	Assigned	Reserved for growth	pwr_base
‘power‘	U2, L1, C1-C5	C6-C9, R5-R9	100 (‘#PWR0100‘+, ‘#FLG0100‘+)
‘mcu‘	U1, Y1, J2, D1, SW1, R1-R4, C10-C19	C20-C29, R10-R19	200
‘usb‘	J1, U3	C30-C39, R30-R39	300

power sheet

Ref	Part	Note
U2	AP63203WU-7, TSOT-26	fixed 3.3 V; FB ties to the +3V3 sense point (NOT a divider - that is the adjustable AP6320x); EN -> VBUS; no PG pin
L1	4.7 uH, Isat >= 1 A, DCR < 100 mohm, shielded	DS41326 Table 2 says 3.9 uH; 4.7 uH is nearer standard and better at light load
C1	10 uF X5R/X7R >= 16 V, 0805	C_IN - the USB-limited one, see power_tree.md s3
C2	100 nF >= 16 V	HF bypass at VIN, tight to the pin
C3	100 nF >= 16 V	C_BST, between BST and SW
C4, C5	22 uF X5R >= 10 V, 0805	C_OUT, ceramic mandatory (internal compensation assumes it)

mcu sheet

Ref	Part	Note
U1	STM32F103C8T6, LQFP-48	
Y1	8 MHz crystal, +/-30..50 ppm, CL 20 pF class	HSE mandatory for USB FS
C10, C11	crystal load caps, C0G	value = 2 x (CL - C_stray); ~22 pF for a CL 20 pF part. No series damping resistor
C12, C13, C14	100 nF	one per VDD/VSS pair (pins 24/36/48)
C15	4.7 uF	+3V3 bulk at the MCU
C16	100 nF	VDDA (pin 9) HF
C17	1 uF	VDDA bulk - kept, not waived: VDDA sits on a 1.1 MHz switching rail
C18	100 nF	VBAT (pin 1)
C19	100 nF	NRST (pin 7)
R1	1 k, 0603	LED series, sets 1.4 mA (PC13's 3 mA limit)
R2	10 k	button pull-up to +3V3
R3	10 k	BOOT0 pull-down to GND
R4	**1.5 k 1%**	USB D+ pull-up to +3V3 - MANDATORY, the F103 has none internally. NOT 10 k (the Blue Pill clone bug)
D1	red LED, 0603	active low
SW1	SMD tactile switch	to GND
J2	1x4 2.54 mm THT pin header	SWD, hand-soldered

usb sheet

Ref	Part	Note
J1	USB micro-B receptacle, SMD	`Connector:USB_B_Micro`; pipeline renumbers `SH` -> 6
U3	USBLC6-2SC6, SOT-23-6	2-channel USB TVS array + VBUS clamp pin

`constraints.json` references J1, J2 (edges), U2/L1/J1/U3 (separation), and Y1/C10/C11 + U2/L1/C1-C5 (groups). `decoupling.json` (P4-emitted) will reference C1, C2, C12-C18 against U1/U2 pins. These assignments are contractual, not suggestions.

3. Interface pinouts (silk-labeled, canonical)

- **J1 micro-B**: 1 VBUS, 2 D- (`/USB_DM`), 3 D+ (`/USB_DP`), 4 ID (**no-connect** - device only, no OTG), 5 GND, 6/SH shield -> **GND directly** (decision 5).
- **J2 SWD**: 1 = +3V3 (reference OUT), 2 = SWCLK, 3 = GND, 4 = SWDIO. ST Nucleo CN4 debug-row order minus NRST/SWO. Silkscreen all four.

4. U1 pin commitments (P4 wiring contract, LQFP-48)

Pin numbers below are from the verified DS5319 pin table used by the stm32-blinky sheet plan plus the USB research (PA11 = 32, PA12 = 33); **P4 re-confirms every number against the 'parts/' datasheet extract** before wiring - PB0's number in particular is not pinned here.

- VBAT (1) -> `+3V3` (no battery) + C18. VDD 24/36/48 -> `+3V3` + C12/C13/C14 + C15 bulk. VDDA (9) -> `+3V3` + C16 + C17. VSS 23/35/47 -> `GND`, VSSA (8) -> `GND`.
- PC13 (2) -> `/mcu/LED` (sink only; ST limits PC13-15 to 3 mA).
- PD0/OSC_IN (5) -> `/mcu/OSC_IN`, PD1/OSC_OUT (6) -> `/mcu/OSC_OUT`.
- NRST (7) -> `/mcu/NRST`.
- PB0 -> `/mcu/BTN` (active low; PA0/WKUP left free for a future active-high wake-up function).
- PA11 (32) -> `/USB_DM`, PA12 (33) -> `/USB_DP`. **No series resistors** - the F103 PHY's output impedance is internal; 22R/33R would push it outside the USB 28-44 ohm window.
- PA13 (34) -> `/mcu/SWDIO`, PA14 (37) -> `/mcu/SWCLK` (SWD is the reset default on these pins).
- BOOT0 (44) -> `/mcu/BOOT0` -> R3 10 k -> GND (fixed strap; the user button is explicitly NOT a boot selector).
- PB2/BOOT1 (20) and every unused GPIO: no-connect flag so ERC is clean.

5. P4 generator notes (schlib specifics that bite)

1. **'power' sheet has no sheet pins**: 'proj.add_sheet(power_sheet.build(), at=..., size=..., nets=[])'. All three of its rails are power symbols and are global already. Do not invent sheet pins for rails. 2. **Use the free-cluster 'hier_pin(net, at=...)' variant** on both the `usb` and `mcu` sheets for `USB_DP`/`USB_DM`. Each net has 2+ components on it, and that variant places a LOCAL label plus the hierarchical label on one stub so the hier label joins by wire geometry rather than relying on hier-label/local-label name merging. 3. **Root-side labels sit 7.62 mm to the LEFT of each sheet pin** (add_sheet's `label_stub = 3 * STUB`), and sheet pins stack down the sheet symbol's LEFT edge. Leave that strip clear when placing the `usb` and `mcu` sheet symbols on the root, and give both sheets the same `nets=` order. 4. **PWR_FLAGs**: `power_flag("VBUS", sym="power:VBUS", flag=True)`, same for `+3V3` and `GND`, all on the `power` sheet. Without them ERC reports the rails as undriven (every driver on this board is a passive pin - connector, inductor). 5. **U3 in-line wiring**: the array's two same-channel I/O pads are internally common - wire BOTH pads of a channel to the same net so layout can pass the trace through the part with no stub. Its VBUS/supply pin ties to `VBUS`. Confirm the pin table from the P3 datasheet extract; do not wire from memory. 6. **Decoupling metadata**: rails on the `mcu` and `power` sheets are power symbols, so the wiring label and the final net name agree (`+3V3`, `GND`) and no `rail_net`/`gnd_net` override is needed anywhere on this board. 7. `Sheet.add_component(..., expect={...})` on U1, U2, U3 - pin-name insurance against a wrong symbol.

6. Board-edge placement (mirrors constraints.json)

- **J1 (micro-B) on the LEFT edge**, pos 0.5 - the plug must seat, and this puts the pair's origin on the short axis with a straight shot to U1 in the middle of the board (usbbuck4 golden precedent).
- **J2 (SWD header) on the RIGHT edge**, pos 0.5 - opposite the USB cable so a debug ribbon and a USB cable do not fight, and PA13/PA14 sit on the LQFP's right flank.

- `separation` pushes U2/L1 at least 8 mm from J1/U3, keeping the 1.1 MHz switch node out of the corner where the USB pair starts.
- `groups`: `xtal` (Y1 + C10/C11) and `buck` (U2 + L1 + C1-C5) so the annealer moves each as a unit until P4's decoupling metadata takes over.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/decisions.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

5 Schematic

Gate `erc` (phase P4): **pass** after 2 attempt(s); last 2026-08-14T11:25:02, failing 0 of 0.

Review waivers

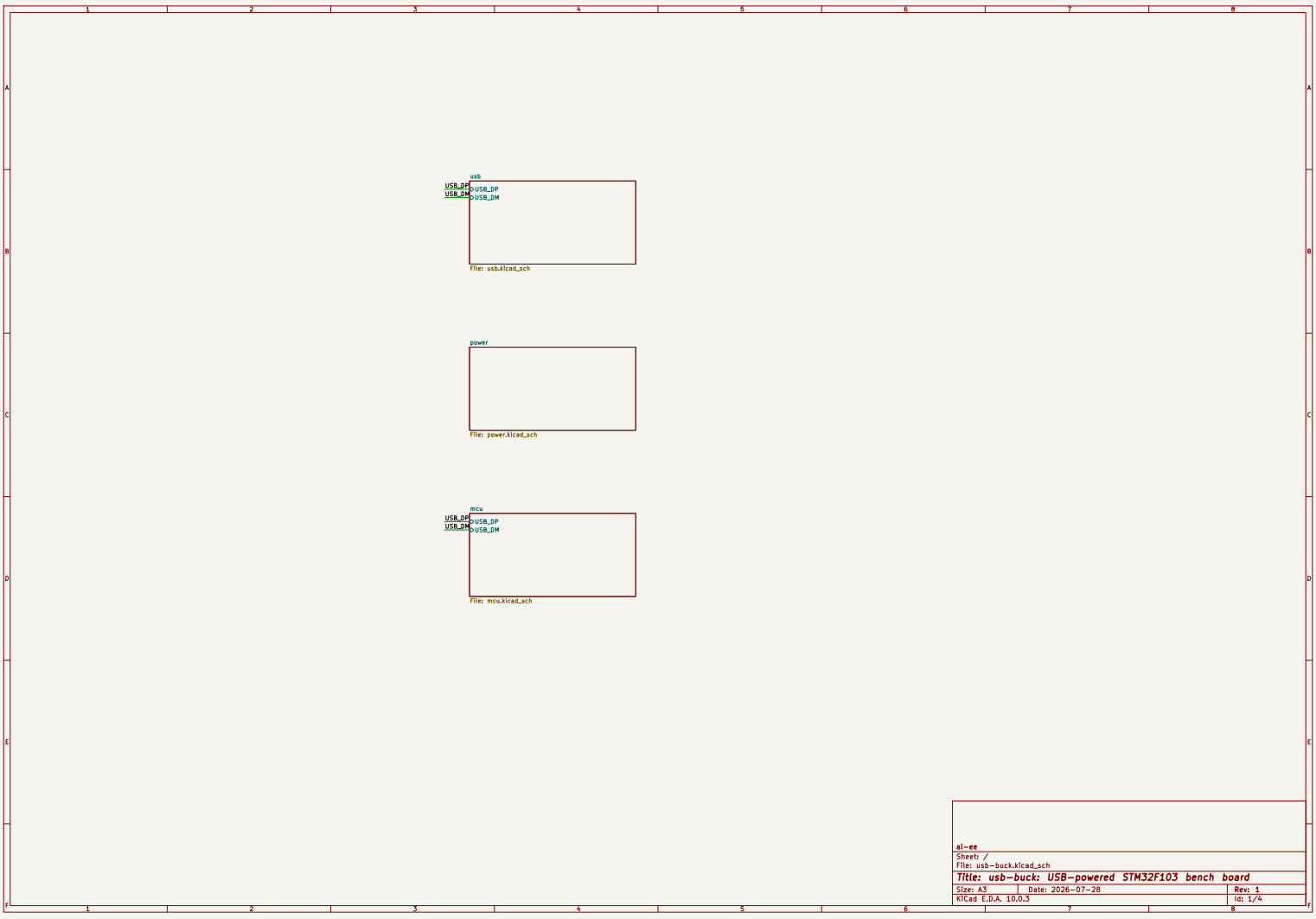
Schematic review waivers (usb-buck)

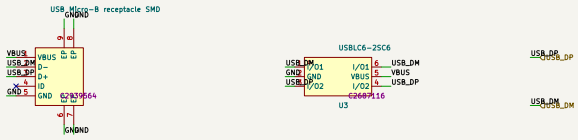
W1 sw1-pairing-unverified: SW1 (C49023761) has no manufacturer datasheet; {1,2}/{3,4} pairing follows the vendor symbol's internal bars. Failure mode if wrong: button reads permanently pressed (feature dead, NO damage; USB/ SWD/LED unaffected). WAIVED with a mandatory bring-up step: continuity-check SW1 pins 1-3 unpressed before firmware relies on the button. W2 usblc6-clone-vrwm: UMW USBLC6-2SC6 rated VRWM 5.0V vs USB legal 5.25V (top 0.25V unspecified). WAIVED for prototype qty 10; ORDER-TIME OPTION: swap to ST original (C7519, rated 5.25V, listed alternate in parts.json) - noted for H5.

P8 board-review waivers (usb-buck)

W3 j1-mouth-recess: micro-B mouth 0.46mm behind board edge (silk-edge clearance tradeoff). Most cables latch at <=0.5mm recess. WAIVED for prototype; respin note: make flush/proud + trim mouth-end footprint silk. W4 economy-smt-tht: J1 4 shield pegs (2x 0.5mm slots) + J2 header arrive UNSOLDERED on economy SMT-only assembly. Mandatory hand-solder step -> in order human_steps at P10. W5 uncoupled-relaxation-scope: max_uncoupled 8.0 is FS-THIS-BOARD-only (structural span; ~1.5mm was recoverable by flipping R4 - respin nicety). Never reuse on HS designs. W6 d1-glyph: interior silk bar contradicts (correct) chamfer; hidden under body post-assembly; rework-doc note only.

The full schematic PDF follows.


usb
USB_DP
USB_DM
USB_DP
USB_DM
File: usb.kicad_sch
power
File: power.kicad_sch
mcu
USB_DP
USB_DM
USB_DP
USB_DM
File: mcu.kicad_sch
al-ee
Sheet: /
File: usb-buck.kicad_sch
Title: usb-buck: USB-powered STM32F103 bench board
Size: A3
Date: 2026-07-28
Rev: 1
KiCad E.D.A. 10.0.3
Id: 1/4





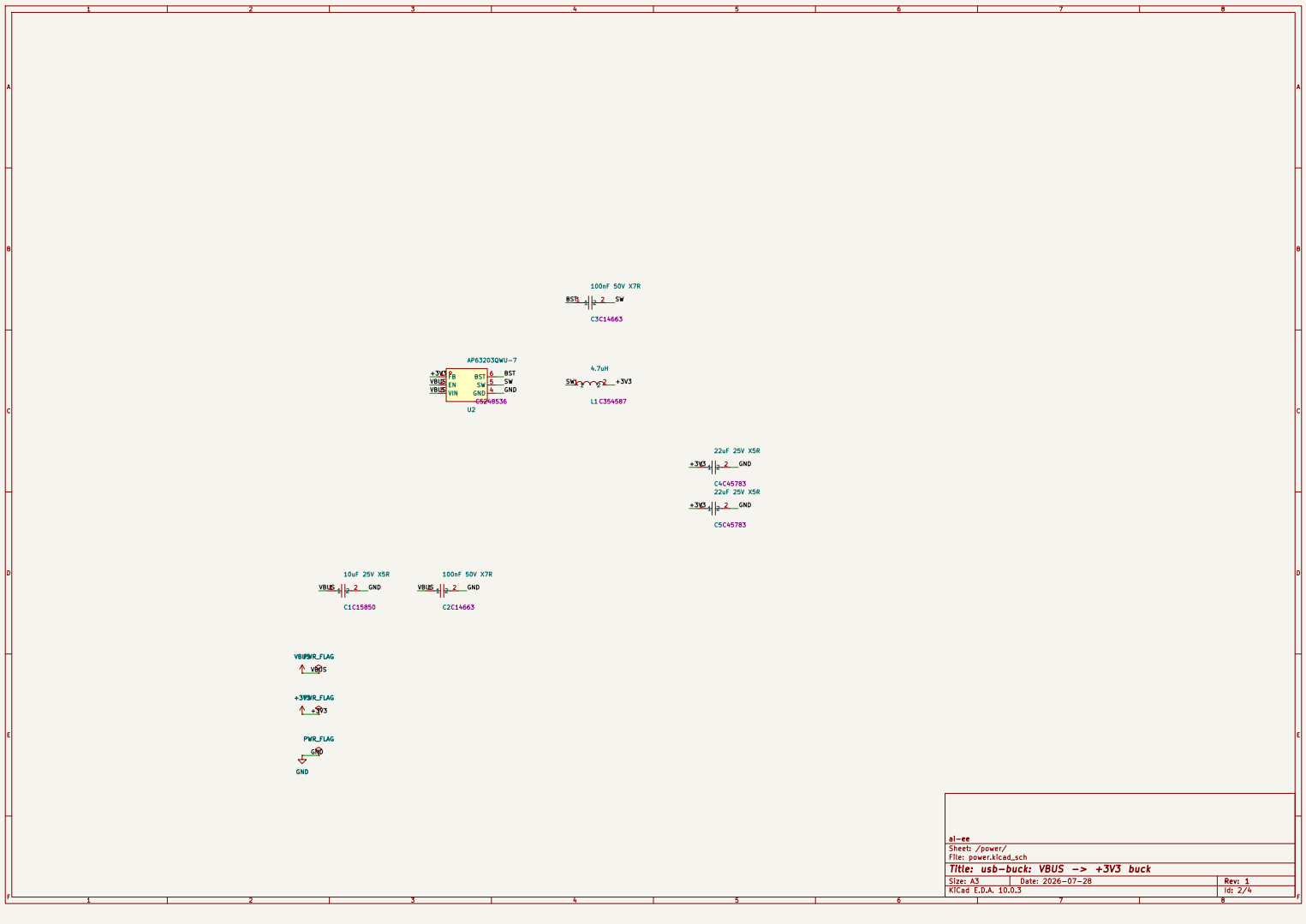
al-ee

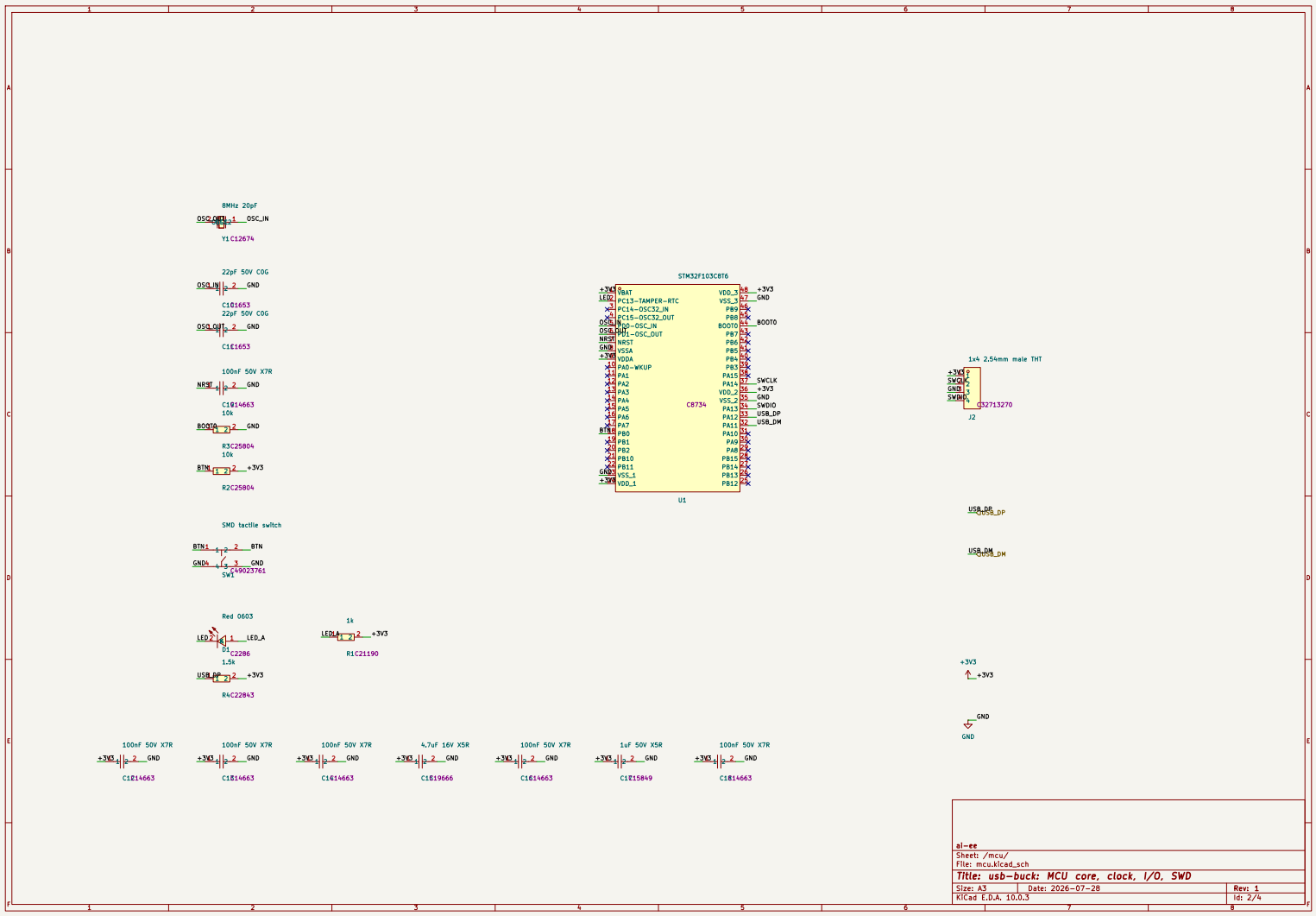
Sheet: /usb/

File: usb.kicad_sch

Title: usb-buck: USB port + ESD

Size: A3	Date: 2026-07-28	Rev: 1
KiCad E.D.A. 10.0.3		Id: 2/4


100nF 50V X7R
BST
SW
C3C14663
AP63203QWU-7
+3V3
VBUS
VBUS
FB
EN
VIN
BST
SW
GND
GND
EXPOSED
C5248536
U2
4.7uH
SW
+3V3
L1 C354587
22uF 25V X5R
+3V3
GND
C4C45783
22uF 25V X5R
+3V3
GND
C5C45783
10uF 25V X5R
VBUS
GND
C1C15850
100nF 50V X7R
VBUS
GND
C2C14663
VBUS
PWR_FLAG
VBUS
+3V3
PWR_FLAG
+3V3
PWR_FLAG
GND
GND
al-ee
Sheet: /power/
File: power.kicad_sch
Title: usb-buck: VBUS -> +3V3 buck
Size: A3
Date: 2026-07-28
Rev: 1
KiCad E.D.A. 10.0.3
Id: 2/4

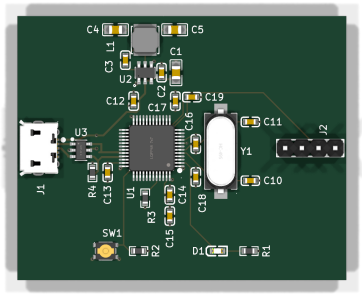


al-ee	
Sheet: /mcu/	
File: mcu.kicad_sch	
Title: usb-buck: MCU core, clock, I/O, SWD	
Size: A3 | Date: 2026-07-28	Rev: 1
KiCad E.D.A. 10.0.3	Id: 2/4

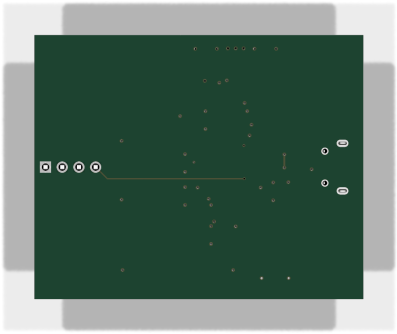
6 Layout

Gate `place` (phase P6): **pass** after 2 attempt(s); last 2026-08-14T11:25:15, failing 0 of 0.

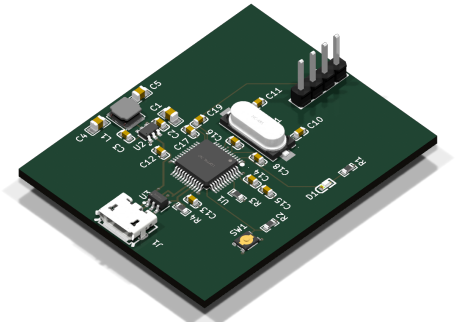
Gate `drc_routed` (phase P7): **pass** after 3 attempt(s); last 2026-08-14T11:25:24, failing 0 of 0.



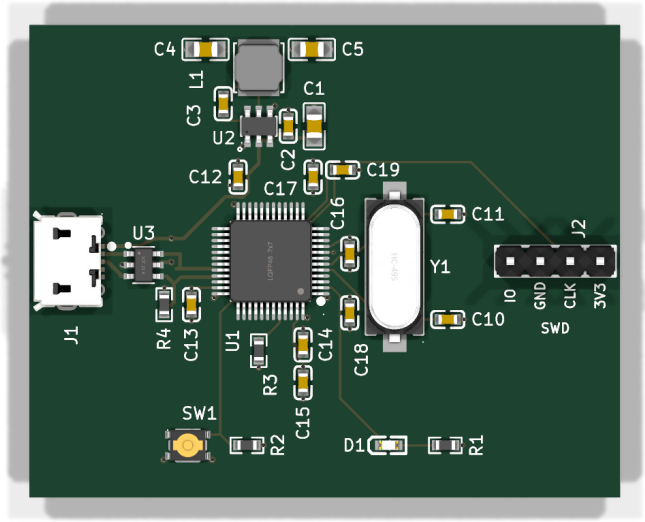
`reports/renders/usb-buck_top.png`



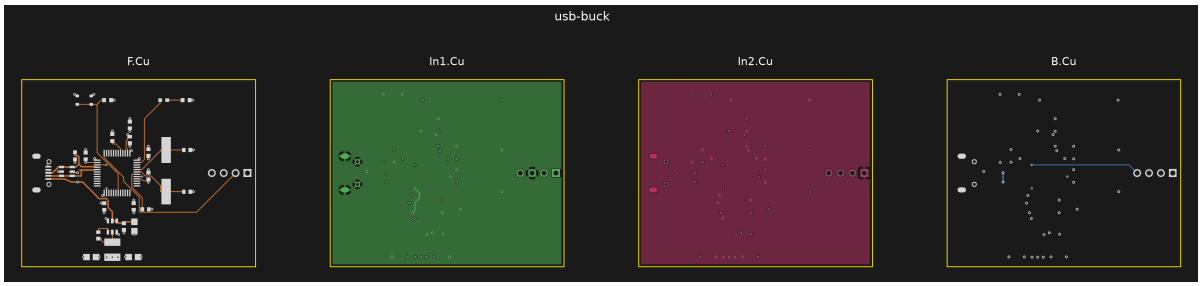
`reports/renders/usb-buck_bottom.png`



`reports/renders/usb-buck_iso.png`



reports/render_labeled/usb-buck_top.png



reports/layers/usb-buck_layers.png

7 Verification

Gate `verify` (phase P8): **pass** after 3 attempt(s); last 2026-08-14T11:25:31, failing 0 of 2. *reports/verify_all.json not available.*

Design review of record

Board review: usb-buck (adversarial, P8)

Reviewer stance: machine gates are green (DRC 0/0 incl. warnings, verify 8/8 checks 0 findings, ERC 0/0); hunt only for what those checks cannot see. Evidence: fresh renders reports/renders/usb-buck_{top,bottom,iso}.png at 2400 px plus close-up crops (crop_usb.png, crop_buck.png, crop_u1.png, crop_right.png, crop_bottomui.png); SWIG geometry interrogation of kicad/usb-buck.kicad_pcb (pads, tracks, vias, zones, silk inventory, collinear-overlap scan); requirements.md, architecture/*, constraints.json, reports/erc-waivers.md, reports/review-schematic.md, kicad/decoupling.json.

Verdict: 1 error, 4 warnings. The electrical design is delivered and the layout fundamentals (buck hot loop, ESD flow-through, plane discipline, decoupling ring) are genuinely good. The error and two of the warnings are "assembly/field reality" defects invisible to every script in the suite.

Findings, worst first

E1 - J2 SWD header has no silkscreen pin labels (error) The board's ONLY silk text is 28 refdes strings. requirements.md s2 promises the SWD pinout is "silkscreen label[ed]"; architecture/blocks.md (Debug) orders "Silkscreen every pin" and explains why: pin 1 carries +3V3 as a

REFERENCE OUTPUT and "powering the board from the debugger would back-drive the buck output". As built (crop_right.png), pin 1 is marked only by a square THT pad - invisible once the header is soldered - and nothing identifies SWCLK/GND/SWDIO. A user with a clone ST-Link (whose 3.3V pin SUPPLIES power) has a 50% chance of connecting the ribbon reversed: SWDIO->3V3 backfeeds the rail, debug fails, and the AP63203 output is back-driven. Minutes to fix (four silk labels + a pin-1 tick), and check_silk can never see it - it checks geometry, not meaning.

W1 - USB mouth recessed ~0.5 mm behind the board edge (warning) J1's mating face sits at board x=11.0 vs edge x=10.54 (courtyard/silk front transformed from the footprint; no F.Fab body outline exists to be more precise). crop_usb.png shows the PCB lip clearly in front of the connector mouth. A micro-B plug must insert ~4.85 mm and compliant plugs expose ~6.5 mm of shell, so ~1.5 mm of slack absorbs this for most cables - but short-shell or chunky-overmold cables will bottom on the lip before the latch engages: intermittent contact, cable falls out. Standard practice is mouth flush to +0.5..1 mm proud. Shift J1 -0.5..-0.8 mm in x at the next spin (the placer respected courtyard-to-edge clearance where a connector wants overhang).

W2 - J1 shield pegs and J2 leave JLC economy assembly unsoldered (warning) JLC economy PCBA is SMT-only. J2 (THT header) is planned as hand-solder per architecture - fine, bottom side is bare and access is excellent (usb-buck_bottom.png). But J1's four shield legs are PTH pads (2x round 0.7 mm, 2x slot 1.3x0.5 mm, all net GND) and NO document calls out soldering them. Fresh from assembly, the shield-to-GND bond (decision 5) and nearly all of the jack's rip-off retention hang on unsoldered pegs in plated holes; the SMD signal pads alone will not survive cable side-loads for long. Failure: shield ground goes intermittent, then the jack tears off. Needs one line in the bring-up/assembly instructions: hand-solder J1's 4 pegs + J2's 4 pins. (The two 0.5 mm-wide slots sit exactly at JLC's minimum slot width - inside capability, zero margin; dfm_check in S12 must confirm.)

W3 - diff-pair constraint relaxation is defensible; its justification is flattering (warning) max_uncoupled_mm 5.0->8.0 was corrected to pass the measured 6.61 mm uncoupled span. Geometry audit: the USBLC6-2SC6 traverse genuinely forces ~3.4-3.6 mm of separation (GND pin 2 sits BETWEEN D-/D+ on the entry side; pads 1.9 mm apart > the 1.56 mm coupling window) - unavoidable with this part, as claimed. The "2.9 mm D+ pull-up stub" measures 3.16 mm junction->R4.1, and ~1.5 mm of it IS avoidable: R4's D+ pad is currently the FAR pad (y38.35); rotating R4 180 deg puts it at y36.85. Additionally the "coupled" remainder runs at 0.71-1.25 mm centre-to-centre against the 0.52 mm 90-ohm geometry (gap_min 0.50 exists only near J1), so blocks.md's "90 ohm pair" is nominal, not delivered. Verdict: ACCEPT for USB full-speed - 12 Mbps / 4 ns edges over a ~12 mm run above an unbroken In1 GND plane makes all of this immaterial (skew 1.93 mm = 13 ps is noise). Keep 8.0 mm for THIS board; do not let this constraints.json travel to any high-speed design, and drop the "zero avoidable meander" framing - R4's orientation was avoidable.

W4 - D1 interior silk glyph contradicts the correct chamfer cathode cue (warning) LED-SMD_L1.6-W0.8-R-RD footprint: the outline chamfer correctly marks the cathode end (pad 2, matches netlist - cathode to PC13 sink, anode via R1 to +3V3; crop_bottomui.png), and it remains visible after assembly. But the interior glyph draws its polarity bar at local x=+0.09 with the tick toward -x - it reads as "cathode at the ANODE end". The glyph hides entirely under the 1.6 mm body once placed, so machine assembly (CPL-driven) and post- assembly inspection are safe. The exposure is hand-rework: with D1 removed, a tech who trusts the interior bar mounts the replacement backwards - LED dead (no damage, diode blocks). Low risk at qty 10; fix the glyph in the footprint at next touch. (fp_verify's 0.8 vs 0.7 mm pad-size warning is a deliberate, benign enlargement - triaged, no action.)

Positively verified (not just "no finding")

- ESD sequencing: J1 (x16.3) -> U3 (x19.1-21.4) -> U1 (x26.75); true flow-through (entry/exit trace runs pad-to-pad under the SOT-23-6 body); U3 GND drops to the In1 plane via at (18.45,34.25), 0.65 mm from pin 2 - exactly the architecture's promise.
- Buck hot loop (crop_buck.png): C2 100nF 1.6 mm from VIN pin 3 with its GND via 0.65 mm away; C1 10uF immediately behind; loop closes into In1 within ~2x2.5 mm. SW node is one 2.15 mm trace to L1 plus the 3.4 mm BST-cap leg - tight and short for a 1.1 MHz sync buck. Output caps C4/C5 flank L1 symmetrically. C2 placement matches the layout note in blocks.md/power_tree.
- Planes: In1 = solid GND, In2 = +3V3, both filled, only 3 tracks on B.Cu (SWDIO run, VBUS underpass) - the underpass's two vias sit 0.7 mm from the D- line but two planes below it; return-path check (radius 2.0) green.
- Crystal: Y1 (HC-49S) is 7.6+ mm from the buck bbox and >16 mm from edges; load caps C10/C11 at the crystal pads with GND vias 0.65 mm out; OSC_IN/ OSC_OUT diverge immediately (no long parallel run); SWCLK passes no closer than ~1.3 mm north of the can. OSC traces are 11.9/12.6 mm - long-ish, the price of the physically huge HC-49S; acceptable at 8 MHz over solid GND.
- Placement contracts: J1/J2 on left/right edge at pos 0.5; buck group vs USB group separation 9.8 mm (>= 8 required); board 50.1 x 40.1 mm inside the 55 x 45 cap; single-sided SMT assembly delivered (bottom bare).
- VBUS after P7 dedup: zero exact duplicates board-wide; VBUS itself also free of partial overlaps. (22 micro-overlaps of 0.2-0.34 mm remain at via junctions on GND/+3V3/D- - source-file cosmetics; copper unions in gerbers, no fab impact.)
- Pin-1 cues: silk dots present and post-assembly-visible for U1, U2, U3; refdes all 1.0 mm, none obscured by parts (renders).
- U1.8/U1.9 escape co-design at (36.3-36.4, 32.9-33.2): VSSA drops to plane at (36.35,33.17), VDDA exits at y32.67 - 0.3 mm edge gap, no sliver.
- Decoupling ring: C12/C13/C14 hug their VDD pins, C16+C17 on VDDA by pin 9, C18 on VBAT, C15 bulk 6.6 mm off pin 48 (fine for bulk), C19 NRST cap 7.5 mm from pin 7 (filter still effective; cosmetic).
- Cross-artifact: every requirements.md interface exists on copper (USB device w/ ESD, SWD, LED, button, crystal, AP63203 buck); every blocks.md block placed; VBUS carries exactly the "only capacitance allowed" set.

Warnings triage (every non-green anywhere gets a verdict)

- Schematic review W1 (SW1 pairing unverified): layout is consistent with the assumed {1,2}/{3,4} pairing (top row = BTN + R2, bottom row = GND vias); if pairing is actually left/right the button reads stuck-pressed - exactly the waived failure mode. Waiver SOUND; bring-up continuity check pins 1-3 remains MANDATORY.
- Schematic review W2 (UMW USBLC6 clone VRWM 5.0 V vs 5.25 V): unchanged by layout; waiver SOUND for qty 10; ST original (C7519) swap remains the order-time option - carry to S12 BOM.
- fp_verify D1 pad-size warning: deliberate 0.05 mm/side enlargement, benign (see W4 for the real D1 issue).

- Historical diffpair error (6.61 > 5.0): superseded by the corrected constraint - judged in W3.
- Skipped checks: none within the P8 suite (8/8 board checks ran). The remaining hole is downstream: dfm_check / BOM-CPL validation (S12) has not run - slot-width minimum, Extended-part status of J1/U2/U3, and CPL rotation for the polarized D1 are all still machine-unverified.

Open (could not judge from renders/geometry alone)

- Whether JLC will place J1 (C2939564, Extended SMD) on economy PCBA and whether any paste-in-hole occurs on its peg holes - S12 territory.
- Actual latch engagement of a worst-case micro-B cable against the 0.5 mm lip (W1) - needs a physical plug test on the v1 article.
- SW1 internal pairing (W1 waiver) - electrical bring-up only.

8 DFM and Fabrication

Gate `dfm` (phase P9): **pass** after 2 attempt(s); last 2026-08-14T11:25:39, failing 0 of 1. DFM findings by severity: warning: 1.

Fabrication outputs

Exported layers: F.Cu, In1.Cu, In2.Cu, B.Cu, F.Silkscreen, B.Silkscreen, F.Mask, B.Mask, F.Paste, B.Paste, Edge.Cuts.

Bill of materials

Comment	Designator	Footprint	LCSC
10uF 25V X5R	C1	C0805	C15850
100nF 50V X7R	C2,C3,C12,C13,C14,C16,C18,C19	C0603	C14663
22uF 25V X5R	C4,C5	C0805	C45783
22pF 50V C0G	C10,C11	C0603	C1653
4.7uF 16V X5R	C15	C0603	C19666
1uF 50V X5R	C17	C0603	C15849
Red 0603	D1	LED-SMD_L1.6-W0.8-R-RD	C2286
USB Micro-B receptacle SMD	J1	USB-SMD_USB-111FD-B-SU	C2939564
1x4 2.54mm male THT	J2	HDR-TH_4P-P2.54-V-M	C32713270
4.7uH	L1	IND-SMD_L4.0-W4.0	C354587
1k	R1	R0603	C21190
10k	R2,R3	R0603	C25804
1.5k	R4	R0603	C22843
SMD tactile switch	SW1	SW-SMD_4P-L3.0-W2.6-P1.80-LS3.4-TL	C49023761
STM32F103C8T6	U1	LQFP-48_L7.0-W7.0-P0.50-LS9.0-BL	C8734
AP63203QWU-7	U2	TSOT-26_L2.9-W1.6-P0.95-LS2.8-BL	C5248536
USBLC6-2SC6	U3	SOT-23-6_L2.9-W1.6-P0.95-LS2.8-BL	C2687116
8MHz 20pF	Y1	CRYSTAL-SMD_L11.5-W4.8-LS12.7	C12674

2 CPL rotation correction(s) applied; no missing LCSC numbers.

Quote

Estimated total $27.45 for qty 10 (unit $2.75). This is an estimate from published pricing; the JLCPCB cart quote page is authoritative.

Human steps before ordering

1. HAND-SOLDER after PCBA arrival: J1 4 shield pegs (2x 0.5mm slots) + J2 SWD header - economy assembly is SMT-only (P8 waiver W4). Also: bring-up continuity check SW1 pins 1-3 unpressed (waiver W1); optional order-time swap U3 -> ST C7519 (waiver W2).
2. Upload boards\usb-buck\fab\usb-buck_gerbers.zip to https://jlcdfm.com/ and review the DFM report (V6: no public API).
3. Upload the same zip at https://cart.jlcpcb.com/quote and confirm the real price against the estimate above.
4. If assembling: upload BOM.csv and CPL.csv, then CHECK THE RENDERED PART PREVIEW for every polarized part (LED/diode/electrolytic) - rotation corrections are the classic failure mode.
5. Review, then pay. Payment is always the human's action.

9 Run Record (Appendix)

Phase digests

P0

```
# P0 digest (usb-buck)
Brief -> requirements.md.  USB-FS device (micro-B, also power), AP63203 buck
5V->3.3V, STM32F103C8T6, LED+button, SWD. 4L, 55x45 max, qty 10, Basic pref.
No safety flags (5V max, <3A). 2 OPEN questions auto-answered w/ defaults
(no cost cap; bare board).  0 human interactions.
```

P1

```
# P1 digest (usb-buck)
Roster:  interface-spec(USB-FS) + power-architect, both ran (parallel, opus).
USB: 90R diff on JLC04161H-3313 (w0.314/s0.2104), gap_mm=0.52 PITCH, t_rise
4ns but explicit return_via_radius 2.0 (4ns radius=120mm would be vacuous),
external 1.5k DP pull-up MANDATORY (F103 has none), NO series R (AN4879),
USBLC6-2SC6 ESD at connector, shell->GND direct (proposed), fragment
smoke-tested against rules_gen/check_return_path/check_diffpair/stitch_vias.
Power:  VBUS(direct) -> +3V3 (AP63203 buck) 57mA peak +30% -> 0.1A declared;
C_IN capped 10uF (USB inrush 7.2.4.1); no thermal flags; buck-vs-LDO stated,
buck stands per brief.  Architect to settle:  LED R value, DP pull-up
hardwired-vs-GPIO, VBUS sense omit, shield bond, hierarchy net naming.
```

P2

```
# P2 digest (usb-buck)
5 blocks / 3 sheets (usb, power, mcu) + thin root, ~28 parts.  JLC04161H-3313
4L standard; In1 GND, In2 +3V3.  40x30 target (55x45 cap).  Nets contractual:
bare VBUS/+3V3/GND global, /USB_DP //USB_DM root-labeled, others /<sheet>/N.
All research OPENs settled (buck, 1k LED, hardwired 1.5k, no VBUS sense,
shell direct, 10uF VBUS ceiling).  ~$62-75/10.  H1 AUTO-approved.
Re-opens at P3:  TVS VBUS clamp pin; crystal CL -> load caps; Basic/Extended.
```

P3

```
# P3 digest (usb-buck)
parts.json 18 distinct / 28 refs; 11 Basic / 7 Extended; ~$5.11/board.
Sourcer caught AP63203WU-7 stock=1 -> QWU-7 sibling swap.  Extracts:  C8734
(reused from run a), C5248536 (buck:  FB->VOUT direct, EN float-enable, 4ms
SS confirmed, L 3.3-15uH DCR<30m - waived at 60m w/ math), C2687116 (ESD:
pin5 IS VBUS, flow-through 1+6/3+4 confirmed - architecture ESD Q closed).
Librarian:  18/18 pulled, fp_verify 4/0/2 benign; found easyeda2kicad dup
bug + API rate limit (3 LEARNINGS); 11/12 courtyards < pad field (placelib
fix carries it); silk-on-pad on 4 passive fps + U2 no printable pin-1 ->
approved sanitize in flight.  J1 shield = 4 THT pins 6-9 all "EP" (netlist
must tie all to GND); J1 slots at JLC 0.5mm minimum (P9 note).
```

P4

```
# P4 digest (usb-buck)
3-sheet hierarchy (usb/power/mcu + thin root) from 5 generators; full
delete-rebuild reproducible.  ERC 0/0; netlist_audit pass (28 comps, 16 nets,
9 decoupling associations); contractual net names exact.  Fable reviewer:
0 errors / 2 warnings (SW1 pairing no-datasheet -> bring-up check waiver;
UMW clone VRWM 5.0 vs 5.25 -> order-time ST swap option).  H2 AUTO-approved.
Facts:  hierarchical netlists drop unconnected-* nets (P5 watch); schlib
--pins blind to project libs (finding 22); LCSC fields visible cosmetic.
```

P5 *(no digest recorded)*

P6

```
# P6 digest (usb-buck)
seed 608->496, anneal best 320, chose cand1 + re-floorplan (+12% HPWL traded
for hot loop/USB flow/ergonomics):  J1 mouth off-board (3D-model-derived),
J1->U3 2.8mm flow-through no-cross -> U1 left face; buck loop C2 1.63mm,
L1-SW 2.15; crystal 10.7mm off SW node; SW1/D1 user-reachable.  Silk 47->0
via 55 move_text ops (V17 machinery at scale).  Gate PASS 0/0; full DRC =
75 unconnected only; probe 1.0.  H3 AUTO-approved.
FINDING 24:  placelib FpPad drops per-pad rotation (agent re-derived J1's 7
rotated pads manually) - placelib fix candidate.
```

P7

```
# P7 digest (usb-buck)
4L chain:  planes (In1 GND + In2 +3V3) -> critical (VBUS trunked; +3V3
correctly plane-skipped; DIFF PAIR structurally unroutable by KRT route_diff
- far-apart flow-through terminals peel to a never-run SE follow-up ->
FR routed it) -> stitch 38/42 (+1 via removed:  hole-to-hole vs J1 PTH drill
the floor MISSED) -> route_auto (no wedge, rung1, KRT finish) -> U1.9 escape
pocket fix (FR's own via in the lane; co-designed 0.45/0.2 via relocation)
-> 17 DUPLICATED VBUS segments removed (FR echoed route_critical's copper
through SES - S11-era silent hazard, invisible to DRC as same-net stacks)
-> plane_repair x2 clean -> cleanup SKIPPED on dry-run evidence (would have
removed 2 load-bearing DM segments - V13 on 4L too).  Gate PASS 0/0,
completion 1.0.  Diffpair:  skew+asymmetry fixed, uncoupled 6.61 structural
-> constraint corrected to 8.0 (decision logged).  Return path 0 errors.
```

P8

```
# P8 digest (usb-buck)
verify PASS 8/8 first attempt (post P7 constraint correction).  Fable
```

```
reviewer:  1 err (J2 SWD unlabeled - fixed w/ 5 pin-locked add_text ops;
fixer caught the ORCHESTRATOR's wrong pin map in its brief and labeled from
board authority:  1=3V3 2=SWCLK 3=GND 4=SWDIO Nucleo CN4 order) + 4 warns
waived (W3 mouth recess 0.46mm, W4 THT hand-solder step -> P10, W5
relaxation FS-only scope, W6 D1 glyph).  Verified good:  ESD position, hot
loop 2x2.5mm, VBUS dedup clean, crystal clear of SW. H4 AUTO-approved.
```

P9

```
# P9 digest (usb-buck)
fab_export 13 files 4L; BOM complete FIRST TRY (board LCSC fields - run (a)
fix paying off), 2 CPL rotation corrections.  dfm a1 FAIL 5x annular:  ALL
false - check measured holes against tangent-overlapping NEIGHBOUR pads
(stitch vias bite their pads by design) -> union-of-containing-flashes fix
+ test (60 fab tests green).  dfm a2 PASS 0/1 warning.
```

P10

```
# P10 digest (usb-buck)
Quote qty 10 assembled 4L 50x40 + order manifest ready_for_human; W1/W2/W4
waiver actions folded into human_steps (hand-solder J1 pegs + J2, SW1
continuity, optional U3->ST swap).  H5 AUTO-approved-for-package.
RUN (b) COMPLETE: brief -> order-ready, 0 mid-run human interactions
(full-auto per directive), all 5 gates green, 8 script/check defects
found+fixed in-run, 3-sheet hierarchy + 4L + diff pair + buck exercised.
```

Gate attempt history

Timestamp	Gate	Status	Failing/Total
2026-07-28T03:19:44	erc	pass	0/0
2026-07-28T04:18:31	place	pass	0/0
2026-07-28T05:18:19	drc_routed	pass	0/0
2026-07-28T05:20:37	drc_routed	pass	0/0
2026-07-28T05:21:14	verify	pass	0/0
2026-07-28T05:51:50	verify	pass	0/0
2026-07-28T05:56:19	dfm	pass	0/1
2026-08-14T11:25:02	erc	pass	0/0
2026-08-14T11:25:15	place	pass	0/0
2026-08-14T11:25:24	drc_routed	pass	0/0
2026-08-14T11:25:31	verify	pass	0/2
2026-08-14T11:25:39	dfm	pass	0/1

Issues

(no entries)

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-07-28T02:13:09	AUTO-approved per directive: 4L 40x30, ~$62-75/10 assembled, risks = sheet-name coupling (deliberate) + TVS VBUS clamp pin (re-opens at P3)

Id	Phase	Status	When	Note
2	P4	approved	2026-07-28T03:32:38	AUTO per directive: ERC 0/0, audit pass, reviewer 0 err/2 warn both waived (erc-waivers.md: SW1 pairing bring-up check; USBLC6 clone VRWM order-time swap option)
3	P6	approved	2026-07-28T04:18:31	AUTO per directive; renders reports/place/render_final/; 3 refdes-legibility nits + J2-right-edge cost noted by agent, accepted
4	P8	approved	2026-07-28T05:51:50	AUTO per directive: verify 8/8 x2, drc_routed 0/0 x3; reviewer 1 err (J2 labels - FIXED via add_text, pin map from BOARD authority) + 4 warn (W3-W6 waived with notes); renders reports/render_labeled + reports/renders
5	P10	approved	2026-07-28T05:56:50	AUTO per directive: order.json ready_for_human; hand-solder + bring-up + swap-option steps folded into human_steps; payment never automated

Generation cost

Not recorded (no reports/cost.json).

10 Artifact Index

File	Size	Note
state.json	20,338 B	
kicad/PCB-0002-A_usb-buck.kicad_pcb	246,376 B	
kicad/PCB-0002-A_usb-buck.kicad_sch	4,376 B	
fab/PCB-0002-A_usb-buck_gerbers.zip	37,517 B	sha256 b25f1a9cd034cedc...
fab/order.json	2,506 B	
fab/BOM.csv	797 B	
fab/CPL.csv	943 B	
brief/brief.md	558 B	
requirements.md	4,740 B	
architecture/blocks.md	5,068 B	
architecture/constraints.json	2,059 B	
architecture/decisions.md	8,401 B	
architecture/power_tree.md	7,978 B	
architecture/sheets.md	10,085 B	
architecture/stackup.md	4,790 B	
reports/board_init.json	23,933 B	
reports/bom_cpl.json	15,203 B	
reports/dfm_release.json	2,115 B	
reports/erc-waivers.md	1,351 B	
reports/fab_export.json	2,501 B	
reports/fp_land_C0603.json	605 B	
reports/fp_land_C0805.json	605 B	
reports/fp_land_J1_C2939564.json	1,320 B	
reports/fp_land_LED_C2286.json	946 B	
reports/fp_land_R0603.json	603 B	
reports/fp_verify_C0603.json	928 B	
reports/fp_verify_C0805.json	930 B	

File	Size	Note
reports/fp_verify_D1_C2286.json	1,752 B	
reports/fp_verify_J1_C2939564.json	1,484 B	
reports/fp_verify_R0603.json	931 B	
reports/fp_verify_U1_C8734.json	1,679 B	
reports/fp_verify_U2_C5248536.json	2,131 B	
reports/fp_verify_U3_C2687116.json	1,299 B	
reports/gate-dfm.json	1,014 B	
reports/gate-drc_routed.json	378 B	
reports/gate-erc.json	371 B	
reports/gate-place.json	842 B	
reports/gate-verify.json	1,396 B	
reports/lib_pull_pass1.json	4,267 B	
reports/lib_pull_pass2.json	3,407 B	
reports/lib_pull_ratelimited.json	5,632 B	
reports/place_metrics_shelf.json	12,716 B	
reports/review-board.json	3,540 B	
reports/review-board.md	9,362 B	
reports/review-schematic.json	1,077 B	
reports/review-schematic.md	7,491 B	
reports/rules_gen.json	1,025 B	
reports/schematic.pdf	130,195 B	
reports/top.net	61,244 B	
reports/verify_release.json	5,035 B	